

Sliding Mode Control for Gradient Tracking and Robot Navigation Using Artificial Potential Fields

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Abstract—This paper introduces a sliding mode control strategy for tracking the gradient of an artificial potential field. The control methodology is applicable to fully actuated holonomic robotic systems with n -degrees of freedom. The controller yields exact tracking of the gradient lines and is invariant with respect to parametric uncertainty and disturbances in system dynamics. A detailed case study for mobile robots introduces the equilibrium point placement method for designing harmonic planar potential fields for circular obstacle security zones. Diffeomorph mappings can be utilized for more complex obstacle security zones. The gradient of the harmonic potential field is shown to always lead away from the obstacle security zone and to continuously approach the goal point.

I. INTRODUCTION

ARTIFICIAL potential fields have evolved to a widespread tool for guiding the motion both of n -link robot manipulators and of mobile robots. Pioneered by Khatib [1], robotic applications utilizing artificial potential fields range from obstacle avoidance [2]–[4] to robot navigation [5]–[7] and global path planning [8]–[12]. Hogan's fundamental work [13] established the methodology of artificial potential fields in the general framework of impedance control of mechanical systems.

The basic idea of the artificial potential field method is to construct a potential field with a gradient acting attractive toward the goal point (global minimum) and repellent from obstacles (local maxima). Three different approaches utilizing artificial potential fields for controlling the motion of a robotic system have been studied:

- *Coupling the artificial potential directly into the system dynamics via the control:* In other words, the control forces/torques are applied colinear to the gradient of the potential field, implementing a dynamic relationship between the robot and its environment [13]. Obviously, this approach does not lead to tracking of the gradient. However, safe collision avoidance is guaranteed for potentials tending to infinity at obstacle boundaries, even in the face of bounded actuator resources [7].
- *Accounting for robot dynamics and actuator limitations already in the design of the potential field:* An example is Krogh's *generalized potential field* approach, considering

the *time* needed to reach the goal point or an obstacle rather than the respective *distance* [2]. For control design in [2], a double integrator models a mobile robot, taking into account actuator saturation, but no additional dynamics.

- *Using feedback control for tracking the gradient of the potential field:* Khatib [1] proposed a feedback linearization strategy, requiring exact knowledge of the robot dynamics. The rather restrictive requirement of exact knowledge was relieved by Koditschek's *natural motion* approach for dissipative mechanical systems [14], [15], extending the philosophy of PD control to nonlinear mechanical systems. The idea is to design a navigation function [7] and to track its gradient depending implicitly on the system inertia. A related strategy utilizes the total energy of the system as the basis for control [16]. However, application of the *natural motion* approach is limited to dissipative mechanical systems not subjected to external forces like gravity.

We propose a different interpretation of an artificial potential field: The gradient is taken as the desired velocity vector field for the robot. The control forces/torques are employed to orient the velocity vector of the robot colinear to the gradient, which results in exact tracking of the desired trajectories delineated by the gradient lines.¹ Invariance along the gradient lines is achieved via sliding mode control, requiring only knowledge of the gradient and of bounds on the system dynamics. The methodology can be applied to any smooth artificial gradient, in fact to any smooth artificial vector field. The invariance property of the sliding mode control algorithm allows true decoupling of the tasks "design of the potential field" and "tracking the gradient".

The exposition of the dynamic model of a fully actuated holonomic robot with n degrees of freedom in Section II is succeeded by the sliding mode control development in Section III. A detailed case study for mobile robots in the plane in Section IV introduces the *equilibrium point placement method* for harmonic potential fields [12]. Following the designed gradient invariantly, the mobile robot continuously approaches the goal point and does not enter the obstacle security zones. The methodology of the sliding mode control algorithm in combination with the equilibrium point placement method are illustrated in several numerical examples.

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¹The term "gradient line" is used to denote state-space trajectories obtained by integration along the gradient of an artificial potential field. It should be noted that it is not necessary to compute the gradient lines explicitly.

II. SYSTEM DYNAMICS AND ARTIFICIAL POTENTIAL FIELDS

In this preliminary section, we specify the class of the robotic systems and artificial potential fields the control design in the subsequent section is valid for. The general idea of control design is applicable to holonomic mechanical systems of arbitrary dimension, described by the general dynamic model

$$M(x)\ddot{x} + f(x, \dot{x}) = u. \quad (1)$$

where $x \in \mathbb{R}^n$ describes the robot configuration (for example joint displacements for manipulators; and location with respect to a fixed reference coordinate system for mobile robots), $M(x) \in \mathbb{R}^{n \times n}$ is the positive definite inertia or mass matrix, $f(x, \dot{x}) \in \mathbb{R}^n$ comprises centripetal, Coriolis, gravitational effects, and additive disturbances, and $u \in \mathbb{R}^n$ are mechanical control forces supplied by ideal actuators. The above model (1) comprises fully actuated n -link robot manipulators and omni-directional "free-flying" mobile robots. Nonholonomic kinematics of mobile robots with tricycle kinematics require special treatment being beyond the scope of this paper.

The terms $M(x)$ and $f(x, \dot{x})$ are taken to be unknown. However, we assume that the Euclidean norm, denoted by $\|\cdot\|$ throughout the paper, of the term $f(x, \dot{x})$ is uniformly bounded over the range of operating conditions $\mathcal{W}$ by a known scalar constant

$$\|f(x, \dot{x})\| \leq \bar{f}, \quad \forall (x, \dot{x}) \in \mathcal{W} \quad (2)$$

and that the positive definite symmetric inertia/mass matrix $M(x)$ is a uniformly bounded function of the configuration vector $x \in \mathbb{R}^n$ [17] as shown

$$\underline{m}\|y\|^2 \leq y^T M(x)y \leq \bar{m}(x)\|y\|^2, \quad \forall y \in \mathbb{R}^n. \quad (3)$$

where $\underline{m}$ and $\bar{m}$ are positive scalar constants. Furthermore, we assume that the system states $(x, \dot{x})$ are available for feedback.

The goal of control is to track the gradient of an artificial potential field as frequently employed in path planning and collision avoidance for robotic systems. To this end, the derivation of the artificial potential field $\mathcal{U}(x)$ is not specified. However, the gradient to be tracked, given as $\mathcal{E}(x) = -\text{grad } \mathcal{U}$, is assumed to be sufficiently smooth. In particular, $\mathcal{U}$ is required to be a C^2 -function with respect to all components of the system configuration vector $x \in \mathbb{R}^n$. The gradient lines designated by the state-space solutions of $\mathcal{E}(x) = -\text{grad } \mathcal{U}$ are taken as the desired trajectories without being explicitly derived.

III. GRADIENT TRACKING CONTROL BASED ON SLIDING MODE

In this section we design a sliding mode controller that forces the mechanical system (1) to track the smooth gradient of an artificial potential field $\mathcal{U}(x)$. Sliding mode control and variable structure control are widely used in linear and nonlinear systems, in particular in robotics. The basic idea is to force a dynamic system $\dot{\mathcal{X}} = f(\mathcal{X}, u)$ to restrict its motion to a manifold called the "sliding manifold" $s(\mathcal{X}) = 0$. This is achieved by directing the system trajectories toward this

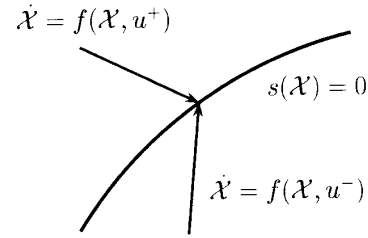


Fig. 1. State vectors in the vicinity of the sliding surface.

manifold "from both sides" using two different controls u^+ and u^- , as shown in Fig. 1.

The main benefits of sliding mode control are its invariance property and the ability to decouple high dimensional problems into sub-tasks of lower dimensionality. The interested reader is referred to DeCarlo's tutorial introduction [18] and Utkin's recent book [19] for a more detailed discussion on sliding mode control. The first application of sliding mode control to trajectory tracking of robot manipulators was presented by Young [20], with subsequent experimental validation by Harashima [21].

In this paper, we exploit the invariance property of sliding mode control for tracking the gradient of an artificial potential field $\mathcal{U}$. In order to guarantee safe operation, exact tracking of the gradient is vital in applications of artificial potential fields like obstacles avoidance. Furthermore, the invariance property allows design of a suitable gradient field independently of the robot dynamics.

In contrast to the approaches described in the introductory section, we choose to regard the gradient lines as the desired trajectories. The gradient is interpreted as a desired velocity field and the controller orients the robot velocity vector co-linear to the gradient. The proposed sliding mode controller provides exact tracking of the gradient lines, requiring only knowledge of the gradient vector, but not of the explicit equation of the gradient lines obtained by integration along the gradient. The controller is robust with respect to system uncertainties and disturbances in the motion equation (1).

The key idea is to regard the velocity vector rather than the acceleration vector as the variable under control. An n -dimensional sliding manifold [19] is defined as

$$s = \dot{x} - v_d(t) \frac{\mathcal{E}}{\|\mathcal{E}\|} = 0, \quad (4)$$

where $\mathcal{E}(x)$ is the gradient to be tracked, and $v_d(t)$ is the scalar desired velocity along the gradient. The desired velocity should be zero at initial and final points, and the associated acceleration should be bounded by a constant scalar $a_0 \geq \max(|\dot{v}_d(t)|)$. A suitable choice for the desired velocity is given by

$$v_d(t) = \min \left(a_0 t, v_0, (2a_0 d(t))^{\frac{1}{2}} \right), \quad (5)$$

where v_0 is the maximal velocity, and $d(t)$ is Euclidian norm of the remaining distance to the goal point.

Restriction of system (1) to the manifold (4) implies that the velocity vector $\dot{x}$ is directed co-linear to the gradient

$\mathcal{E} = -\text{grad } \mathcal{U}$ with its magnitude being specified by $v_d(t)$. Consequently, the gradient lines are tracked exactly.

Bounded maximal velocity v_0 and acceleration a_0 , and normalization of the gradient guarantee boundedness of control. Starting from the initial time $t = 0$, the desired velocity is linearly increased using maximal acceleration a_0 , until the desired “traveling” velocity v_0 is reached. The choice $v_d(t) = (2a_0d(t))^{\frac{1}{2}}$ during the approaching phase prevents “overshoot” at the goal point and results in a finite approach, in contrast to asymptotic approach as in [14], [15]. Provided sliding mode occurs, the robot velocity vector will be colinear to $\mathcal{E} = -\text{grad } \mathcal{U}$ in a vicinity of the goal point. Therefore, motion of the system (1) is restricted to the manifold (4) and can be reduced to a description of the scalar distance to the goal point [10]:

$$\dot{d}(t) = -(2a_0d(t))^{\frac{1}{2}}. \quad (6)$$

The solution of (6)

$$d(t) = \left(-\left(\frac{a_0}{2}\right)^{\frac{1}{2}}t + d(0)^{\frac{1}{2}} \right)^2 \quad (7)$$

vanishes after a finite time interval:

$$d(t) \equiv 0 \text{ for } t \geq \left(\frac{2}{a_0}d(0) \right)^{\frac{1}{2}}. \quad (8)$$

Given the bounds on $f(x, \dot{x})$ in (2) and $M(x)$ in (3), and the bounds on desired velocity and acceleration, v_0 and a_0 , a standard approach is to enforce sliding mode in the manifold $s = 0$ (4) in the domain where

$$\frac{d}{dt} \left(\frac{\mathcal{E}}{\|\mathcal{E}\|} \right) = \frac{\partial}{\partial x} \left(\frac{\mathcal{E}}{\|\mathcal{E}\|} \right) \dot{x} \quad (9)$$

is bounded by using the “unit” control [22], [23]

$$u = -u_0 \frac{s}{\|s\|}, \quad (10)$$

with u_0 denoting the available control force. For arbitrary actuator constraints with different control resources in each of the coordinates of $u \in \mathbb{R}^n$, the method presented in [24] can be utilized.

To ensure the existence of sliding mode in the manifold $s = 0$ given in (4), we examine the *Lyapunov function* candidate

$$V = \frac{1}{2} s^T s. \quad (11)$$

Differentiating (11) along the motion trajectories (1) yields

$$\dot{V} = s^T (M(x)^{-1}u + F(t)), \quad (12)$$

where

$$F(t) = -M(x)^{-1}f(x, \dot{x}) - \dot{v}_d(t) \frac{\mathcal{E}}{\|\mathcal{E}\|} - v_d(t) \frac{d}{dt} \left(\frac{\mathcal{E}}{\|\mathcal{E}\|} \right). \quad (13)$$

Substituting (10) into (12) results in

$$\dot{V} \leq \|s\| \left(-\frac{u_0}{m} + \|F\| \right). \quad (14)$$

It follows from (14) that

$$\dot{V} < -\xi V^{\frac{1}{2}} \quad (15)$$

for some scalar $\xi > 0$ and some finite value of u_0 with

$$u_0 > \xi + \bar{m} \max(\|F\|). \quad (16)$$

Equation (15) and condition (16) imply convergence to the manifold $s = 0$. The time interval needed for convergence is *finite* and can be determined by direct solution of (15), following a similar line of argument as in (6)–(8). It is important to note that for zero initial velocity and bounded desired acceleration a_0 , sliding mode occurs *immediately* without a transient phase for appropriately chosen $v_d(t)$. Consequently, the bounded control (10) leads to ideal tracking of the desired trajectories prescribed by the gradient lines of the potential function $\mathcal{U}$.

A crucial issue are the boundedness assumptions in (9). Although $\frac{\mathcal{E}}{\|\mathcal{E}\|}$ is bounded, $\frac{d}{dt} \left(\frac{\mathcal{E}}{\|\mathcal{E}\|} \right) = \frac{\partial}{\partial x} \left(\frac{\mathcal{E}}{\|\mathcal{E}\|} \right) \dot{x}$ may become unbounded. Physically, this means that the curvature of the gradient lines should be bounded to enable tracking with finite control forces/torques, which are equivalent to finite acceleration.

For real-life control design with specified maximal control resources u_0 , the above procedure is followed in inverse order. Given the maximal curvature $\frac{\partial}{\partial x} \left(\frac{\mathcal{E}}{\|\mathcal{E}\|} \right)$ of the gradient lines and the bounds on the robot dynamics (2) and (3), maximum possible velocity and acceleration are determined to fulfilled condition (16) to guarantee stability in the manifold $s = 0$ (4).

The unit control (10) enforces sliding mode in the origin of the n -dimensional sliding manifold $s = 0$, i.e. in the intersection of the surfaces $s_i = 0$, $i = 1, \dots, n$. The control forces/torques defined in (10) hence are discontinuous only at the origin $s = 0$. Discontinuous control forces/torques are not feasible with real-life actuators due to their inertia. Neglected actuator dynamics and other unmodeled system dynamics may trigger finite-frequency oscillations referred to as “chattering” in nonlinear control literature. An efficient tool to overcome the chattering problem is to employ asymptotic state observers [25]. The key idea is to design a state observer for $(x, \dot{x})$ and to enforce sliding mode in the observer system. Utilization of the observed states $(\hat{x}, \hat{\dot{x}})$ in the sliding manifold (4) generates an auxiliary observer loop *without* unmodeled dynamics, resulting in ideal sliding mode within the observer loop and in chattering-free motion in the actual system. The performance of the actual system depends on the “quality” of the observer.

In many cases, the velocity vector $\dot{x}$ is not measurable, in particular in robot manipulators. Since numerical differentiation of the configuration vector x is undesirable, the introduction of state observers for $(x, \dot{x})$ using x as an input simultaneously solves this problem.

IV. CASE STUDY: MOBILE ROBOT IN THE PLANE

This section is dedicated to the design of a planar potential function for navigation of mobile robots in a workspace $\mathcal{W}$ with known obstacles. One of the primary concerns in the

design of artificial potential fields are local minima causing premature termination of the operation [9], [26], [27], as experienced for example in [1], [3]. Attempts to eliminate local minima in planar potential fields were discussed in [5] and include *navigation functions* [6], [7], *elliptical* artificial potentials [8] and *superquadratic* [9] artificial potentials. These approaches gradually transform the shapes of attractive and repulsive potentials in order to construct a combinational potential field without the generation of local minima, or at least with only “spurious minima” [9]. It was shown by Rimón and Koditschek [5], [7] that there always exists a suitable navigation function for a class of “star-shaped”² obstacle security zones. Using diffeomorph mappings, the “star-shapes” are transformed into discs in a “model sphere world” in which the navigation function then is constructed [7].

In the planar case, the design of a suitable potential field is significantly simplified by the works of Rimón and Koditschek [7]. Given the set of diffeomorph mappings of star-shaped obstacle security zones into circular security zones, it suffices to concentrate on constructing a potential field for circular security zones. The gradient of the potential field to be designed should have the following properties:

Property 1: None of the desired trajectories delineated by the gradient lines should cross into the obstacle security circle.

Property 2: The desired trajectories delineated by the gradient lines should be “reasonably” short.

Property 3: The curvature of the gradient lines should be bounded to enable tracking with finite control resources while maintaining some constant velocity.

Property 4: The potential field, and its gradient, respectively, should be easy to compute with regard to on-line implementation.

The following design of a planar potential field is based on a simple model for a mobile robot, a point mass representing an omni-directional “free-flying” robot. The method can also be applied to mobile robots with nonholonomic kinematics since planning is performed in 2D world-frame $\mathcal{W}$. However, the gradient tracking control algorithm has to be modified adequately to take into account the robot kinematics.

We now summarize the results presented in [12] for global planning in workspaces with known obstacles. The proposed *equilibrium point placement method* has also been successfully applied to sensor-based local navigation of mobile robots operating in unknown environments [28]. Special emphasis is placed on solving the problems Koren and Borenstein described to be inherent to the potential field method [27].

Basic Configuration: One Obstacle

In order to eliminate local minima, we employ harmonic potential fields satisfying Laplace’s equation $\nabla^T \nabla \mathcal{U} = 0$. Harmonic potential fields by nature exhibit no stable equilibrium points away from singular points. Physical interpretations of harmonic potentials can be found for example in electrostatics [10]–[12] and in dynamics of incompressible fluids [4]. For

² A “star-shaped” set has a distinct center point from which all rays cross its boundary once and only once [7].

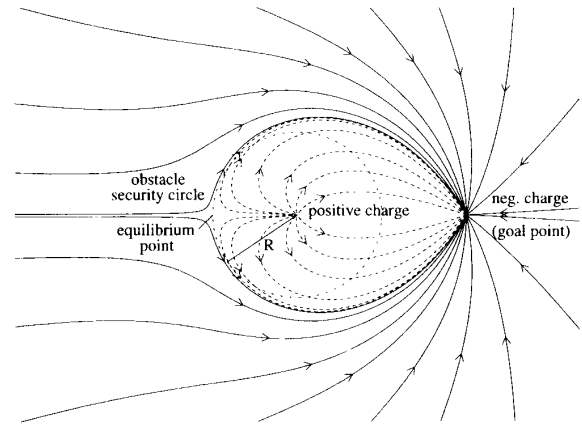


Fig. 2. Some gradient lines of an electrical potential field consisting of a positive charge in the security circle center and a negative charge in the goal point.

convenience, we will use electrostatics’ terminology in the sequel, denoting singular points of the potential field as “point charges”. In planar Euclidean space (x, y) , the harmonic potential of a point charge q is given by

$$\mathcal{U} = q \ln \frac{1}{r}, \quad r = (x^2 + y^2)^{\frac{1}{2}}, \quad (17)$$

with an associated gradient (or electrostatic “force field”)

$$\mathcal{E} = -\text{grad } \mathcal{U} = \frac{q}{r^2} \begin{bmatrix} x \\ y \end{bmatrix}. \quad (18)$$

For each obstacle security circle with radius R , the potential field is designed separately to avoid a mathematical programming problem in the case of multiple obstacles [11]. The *unstable* equilibrium point of the harmonic potential field is placed on the boundary of the security circle R on the side opposite of the goal point. Some gradient lines of the potential field of a negative unit charge in the goal point and a positive charge $q = \frac{R}{R+D}$ in the circle center at distance D from the goal point are shown in Fig. 2.

The following theorems illustrate compliance of the gradient field with the desired Properties 1 and 2. The proofs are sketched in the Appendix, and can be found in detail in [12].

Theorem 1: “Not entering the obstacle security circle.” Consider a harmonic potential field (17) consisting of one negative unit singular point and one positive singular point of strength

$$q = \frac{R}{R+D}, \quad (19)$$

where $R > 0$ is the radius of a circle around the positive singular point and $D > 0$ is the distance between the two singular points. The gradient lines of the harmonic potential (18) only cross the circle boundary from inside to outside (Property 1).

Theorem 2: “Toward the goal point.” Consider a harmonic potential field (17) consisting of one negative unit singular

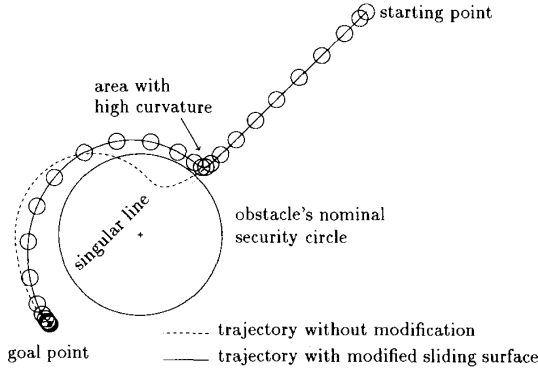


Fig. 3. Motion trajectories without modification and with modified sliding surface.

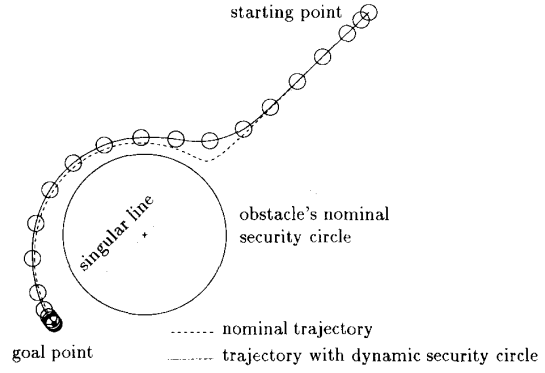


Fig. 4. Nominal trajectory and motion with dynamic security circle modification.

point and one positive singular point of strength q (19), as described in Theorem 1. All gradient lines of the harmonic potential (18) outside the circle R around the positive singular point continuously approach the negative singular point (Property 2).

Theorem 3: “Invariance under coordinate transformations.” Properties 1 and 2 of the planar gradient of the harmonic potential field described in Theorems 1 and 2 are invariant under linear coordinate transformations.

Property 3 is not fulfilled since it can be seen from Fig. 2 that the curvature of the gradient lines tends to infinity close to the equilibrium point. Consequently, tracking of the normalized gradient may not be possible for some desired velocity and with finite control forces. However, only the unstable “singular line” through the two singular points leads directly to the equilibrium point.

A suitable measure to enable continuous tracking of the gradient is to slow down the motion in the area of high curvature by modifying the control algorithm. Redefining the sliding surface (4) as

$$s = \dot{x} - v_d(t) \frac{\mathcal{E}}{\max(\|\mathcal{E}\|, \epsilon_U)} = 0 \quad (20)$$

automatically decreases the velocity in the region where the curvature is high and the absolute value of the gradient is small, see Fig. 3. Provided ϵ_U is chosen such that the boundedness conditions (9) are fulfilled, sliding mode exists along (20), and exact tracking of the gradient lines is guaranteed.

The time needed to reach the goal point might increase considerably for initial conditions very close to the singular line. However, any small perturbation will cause the mobile robot to leave the unstable trajectory along the singular line. In Fig. 3, “numerical perturbations” sufficed to lead the robot away from this trajectory.

The effect of directing the robot away from the singular line can be amplified by heuristic measures. We propose to compensate for the speed with which the robots approaches the obstacle by modifying the radius of the security circle dynamically as follows

$$R_s = \max \left(R - \frac{\dot{x}_r |\dot{x}_r|}{2\epsilon_s a_{max}}, 0 \right), \quad a_{max} = \frac{u_o - \bar{f}}{\sqrt{2\bar{m}}}, \quad (21)$$

where R is the nominal radius of the obstacle security circle being considered, $\dot{x}_r$ represents the radial velocity component directed toward the security circle center (defined negative when approaching the obstacle), $\epsilon_s \in [0, 1]$ is a security factor that should be chosen close to 1, u_o is the available control force (10), $\bar{m}$ is the upper bound on the robot mass (3), and $\bar{f}$ is the upper bound on the system dynamics uncertainty (2). The obstacle charge q in (19) is adjusted according to (21). This heuristic measure is of similar taste as Krogh’s generalized potential field [2], and chooses the obstacle “charge” as high as necessary and as low as possible, as shown in Fig. 4.

Extension: Multiple Obstacles

In the previous section, we discussed the design of a potential field for a basic configuration with a single obstacle security circle of radius R . In order to keep the computational burden low in the case of multiple obstacles (Property 4), only the closest obstacle is taken into account at any time instance. This requires to “switch” the potential field as the robot moves. A precomputed look-up table provides the necessary information about which obstacle should be considered in a respective region, together with the geometric data about the obstacle’s security zone.

The gradient is discontinuous when switching the potential field from one obstacle to the next. Naturally, tracking is not possible at discontinuity points with finite control resources. The sliding mode tracking algorithm utilizes all available control resources to readjust the motion to the new gradient and exact tracking is reestablished after finite time.

The equidistance line between two obstacles security zones forms a “spatial” switching surface in terms of sliding mode terminology, denoted by $s_{sp} = 0$ in Fig. 6(a). Although the gradient lines may be directed toward the spatial switching surface from both sides, tracking is inhibited by the robot inertia. The inertia represents unmodeled dynamics for the spatial switching system, leading to oscillations similar to the oscillations described by Koren and Borenstein [27] for motion in narrow passages, referred to as “chattering” [19] in sliding mode terminology. Fig. 5 depicts an illustrative example.

The chattering behavior can be eliminated by switching the potential field “smoothly” using a continuous approximation

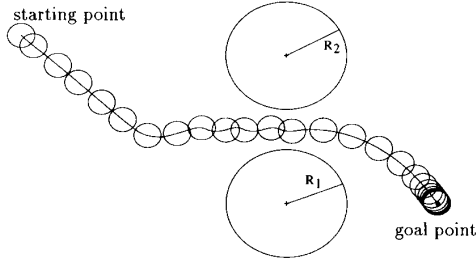


Fig. 5. Oscillatory robot trajectory for discontinuous switching of the gradient between two obstacles.

for the discontinuous control law (10). The approximation of the discontinuity can be defined with respect to time as well as with respect to the distance to the equidistance line s_{sp} . In other words, in a boundary layer δ along the equidistance line between the obstacles, the *two* closest obstacles should contribute to the potential field with time-dependent or state-dependent weights.

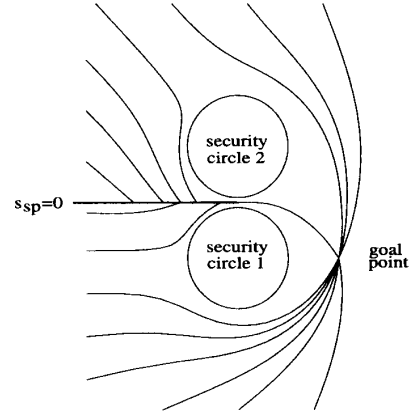
Provided “smooth” switching is defined such that the curvature of the gradient lines does not exceed the maximal acceleration capability of the mobile robot at any point, see Fig. 6(b), the sliding conditions (9) will not be violated when entering the boundary layer δ and sliding mode will continue along (20). This can be established by using the frame work of (11)–(14) and is discussed in detail in [29]. The properties of each gradient designed for a single security zone are preserved for the resultant gradient as long as the width of the boundary layer δ is smaller than the smallest distance between the two security zones, which of course do not necessarily have to be circles as in Fig. 6.

In most previous implementations, time-dependent smoothing was employed, for example using a low pass filter. However, time-dependent smoothing can only damp the oscillations, whereas state-dependent approximation of the discontinuity eliminates the oscillations completely. A possible choice of a state-dependent approximation is

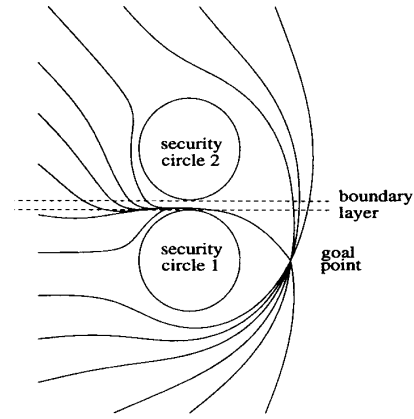
$$\text{grad } \mathcal{U}^* = \begin{cases} \text{grad } \mathcal{U}_1 & \text{for } s_{sp} < -\frac{\delta}{2} \\ \mu \text{ grad } \mathcal{U}_1 + (1-\mu) \text{ grad } \mathcal{U}_2 & \text{for } |s_{sp}| < \frac{\delta}{2} \\ \text{grad } \mathcal{U}_2 & \text{for } s_{sp} > \frac{\delta}{2} \end{cases} \quad (22)$$

where s_{sp} is the distances to the equidistance line $s_{sp} = 0$ (counted positive on the “side” of obstacle 2), and $\mu = \frac{1}{2} + \frac{s_{sp}}{\delta}$. $\mathcal{U}^*$ has a bounded derivative, which ensures that “time” sliding mode exists along (20). Navigation through a narrow passage between two obstacles can be achieved without oscillations. The resulting trajectory for the previous example is shown in Fig. 7.

Fig. 8 shows the motion trajectory of a circular robot in an environment with three obstacles approximated by ellipses. Different ellipse-to-circle transformations were utilized for each obstacle.



(a)



(b)

Fig. 6. (a) Gradient lines for discontinuous gradient switching; (b) Gradient lines for smooth gradient switching.

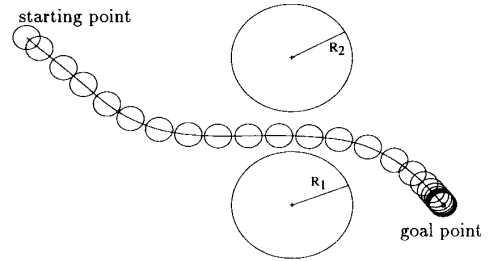


Fig. 7. Motion trajectory in arrow passage using a stage-dependent approximation for gradient switching.

V. CONCLUSIONS

In this paper, we introduced a sliding mode controller for fully actuated holonomic robotic systems with n degrees of freedom. The controller guaranteed exact tracking of the gradient lines of an artificial potential field. The key idea was to orient the robot velocity vector colinear to the gradient. The

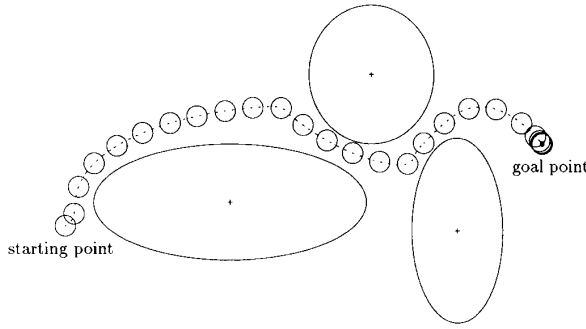


Fig. 8. Motion trajectory in an environment with three obstacles.

motion trajectories were shown to invariantly coincide with the gradient lines in spite of uncertainties in the system dynamics. The invariance property opens completely new perspectives for path planning and obstacle avoidance using the potential field method both in *global* and *local* problems. Since the sliding mode tracking control algorithm is independent of the particular artificial potential field being used, the design task is simplified to constructing a potential field such that all gradient lines lead to the goal point and do not cross the obstacle boundaries.

The *equivalent point placement method* was presented for designing planar harmonic potentials for mobile robot applications. Previously encountered problems in implementing artificial potential fields like local minima and oscillations in motion trajectories, were eliminated. The methodology was applied to mobile robots operating in an environment with known obstacles. Circular security zones were utilized for each obstacle. Diffeomorph mappings can be employed for more complex obstacle security zones. At most two obstacles were considered at each time instance; still the trajectory continuously approached the goal point. Due to its low computational complexity, the method shows promising perspectives for implementation in unknown and partially unknown environments for sensor-based path control of mobile robots.

VI. APPENDIX

Please refer to [12] for details of the proofs outlined below.

Outline of the Proof of Theorem 1:

The proof of Theorem 1 is most easily performed in a polar coordinate system with its origin being located in the center of the obstacle security circle.

In a vicinity of the positive singular point with strength q given in (19), the radial component of the gradient, $\mathcal{E}_r$, is positive, whereas $\mathcal{E}_\varphi$ is negative "far away" from the origin. Solving

$$\mathcal{E}_r(r, \varphi) = \frac{q}{r} - \frac{r - D \cos \varphi}{r^2 - 2Dr \cos \varphi + D^2} \equiv 0, \quad (23)$$

yields a closed curve that encircles the security circle with radius R around the positive singular point in the origin, testifying to $\mathcal{E}_r > 0$ inside the security circle.

Outline of the Proof of Theorem 2:

The proof of Theorem 2 is based on a polar coordinate system with its origin being located in the goal point. Similarly to the proof of Theorem 1, we examine the loci where $\mathcal{E}_r \equiv 0$ and show that $\mathcal{E}_r < 0$ outside the security circle of radius R around the positive singular point located at distance D from the negative singular point.

Outline of the Proof of Theorem 3:

It suffices to show that the sign of the radial component of a vector is invariant under linear coordinate transformations. For illustration purposes, consider the ellipse-to-circle transformation mapping an ellipse in (x, y) with the axis R_x and R_y into a unit circle in $(\tilde{x}, \tilde{y})$. Clearly, the sign of the radial component of a vector is retained in transformed space $(\tilde{x}, \tilde{y})$.

The proof of invariance of the results of Theorems 1 and 2 under different diffeomorph mappings, for example affine mappings, follows a similar line of argument and is left to the interested reader.

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